

**D.Y. PATIL COLLEGE OF ENGINEERING &
TECHNOLOGY, KOLHAPUR**

(An Autonomous Institute)



DEPARTMENT OF CSE (DATA SCIENCE)

A

Project Report

on

“Real Time Speech to Sign Language Translation”

Submitted by

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Under the guidance of

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Final Year B. Tech. CSE (Data Science)

Academic Year 2024-25

D. Y. PATIL COLLEGE OF ENGINEERING & TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)



DEPARTMENT OF CSE (DATA SCIENCE) CERTIFICATE

This is to certify that,

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have successfully completed the project work entitled,

“Real Time Speech to Sign Language Translation”

In partial fulfilment for the curriculum of **Final Year B. Tech. CSE (Data Science)**. This is the record of their work carried out during academic year 2024-2025.

Date:

Place: Kolhapur

Prof. Dr. G. V. Patil
Project Guide

Prof. Dr. G. V. Patil
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Principal

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DECLARATION

We the undersigned students of **Final Year B. Tech. CSE (Data Science)** declare that the project work report entitled **“Real Time Speech to Sign Language Translation”** written and submitted by us, under the guidance of **Prof. Dr. G. V. Patil** is our original work. The empirical findings in this report are based on the data collected by us. The matter assimilated in this report is not the reproduction of any readymade report. We have not violated any of the provisions under the Copyright and Piracy / Cyber / IPR Act amended from time to time.

Date:

Place: Kolhapur

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Date:

Place: Kolhapur

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ABSTRACT

The proposed system aims to support hearing-impaired individuals by translating real-time spoken English into American Sign Language (ASL) using a web-based solution. Built entirely with JavaScript, the system captures live speech using the SpeechRecognition Web API, processes the audio through AudioContext for extracting features like frequency, amplitude, and spectral information (including spectral centroid and bandwidth). It also utilizes Natural Language Processing (NLP) techniques via external Gemini AI APIs to refine and punctuate the transcribed text, and further analyze the emotional sentiment (positive, neutral, negative) of the spoken content. The processed content is then translated into ASL gestures, which are animated using Framer Motion and dynamically rendered through require.context (Webpack) for displaying sign images. Additional libraries such as Recharts are used for visualizing audio features and sentiment data, Axios for API communication, and Lucide-react for intuitive UI elements. This system aims to make real-time speech more accessible and inclusive, offering a meaningful, interactive communication experience for users with hearing or learning impairments.

INTRODUCTION

a. Need of the Work

Sign language plays a vital role in helping individuals with hearing or speech impairments communicate in their daily lives. Among all sign languages, American Sign Language (ASL) has seen significant development, with many systems and tools already available. However, most of these systems are either limited in functionality or not real-time. There is still a need for accessible, interactive platforms that can translate spoken English into ASL gestures, especially in a web-based environment. This project addresses that gap by using modern web technologies to offer real-time speech-to-ASL translation.

b. Existing System

Existing sign language translation systems commonly follow approaches like Direct Translation, Statistical Machine Translation, or Transfer-based Architecture. Most ASL systems focus on text-to-sign video conversion or avatar animations. However, few support real-time speech input, especially within browsers. Many of them also require complex installations or are restricted to desktop applications. Our work stands out by providing a lightweight, browser-based system that translates spoken English into ASL using live speech recognition and dynamic sign rendering.

c. Proposed System

The proposed system is designed for individuals with hearing impairments who use ASL as their primary language. It addresses the communication barrier between hearing individuals and the deaf community by providing a real-time speech-to-ASL translation platform. Developed using JavaScript and modern front-end libraries, the system captures spoken English through the SpeechRecognition Web API, processes it using NLP techniques with support from external Gemini AI APIs for punctuation correction and sentiment analysis, and then translates the content into ASL gestures. These gestures are animated using Framer Motion and displayed using dynamically loaded sign images via Webpack's `require.context`. It also performs audio feature analysis—extracting frequency, amplitude, and spectral characteristics—visualized through Recharts graphs. This system is valuable for ASL learners, educators, or anyone aiming to improve communication with the ASL community. By combining accessibility, usability, and emotional context detection, the system enables more inclusive and meaningful interactions.

LITERATURE SURVEY

American Sign Language (ASL) has been the focus of numerous technological advancements aimed at facilitating communication for the deaf and hard-of-hearing community. One notable development is DeepASL, a system that utilizes infrared sensors to non-intrusively capture ASL signs. It employs a hierarchical bidirectional deep recurrent neural network (HB-RNN) alongside a Connectionist Temporal Classification (CTC) framework to enable both word-level and sentence-level translation. Evaluations of DeepASL demonstrated an average word-level translation accuracy of 94.5% and an 8.2% word error rate for unseen sentences, highlighting its potential in real-world applications [1].

In addition to vision-based systems, wearable technologies have been explored for ASL translation. The Myo armband, equipped with electromyography (EMG) sensors and inertial measurement units, has been utilized to detect muscle activity and motion for gesture recognition. Studies have shown that combining the Myo armband with machine learning algorithms, such as support vector machines (SVMs), can effectively classify a range of ASL gestures with high accuracy. This approach offers a portable and user-friendly solution for real-time sign language translation [2].

Further research has integrated the Myo armband with additional sensors to enhance ASL recognition capabilities. For instance, combining the Myo armband with devices like the Leap Motion controller allows for the capture of both muscular and spatial data, improving the system's ability to interpret complex gestures. Such multimodal systems have demonstrated improved accuracy and robustness in various environmental conditions, making them suitable for diverse applications [3].

Despite these advancements, challenges remain in achieving seamless and natural ASL translation. Systems must account for the nuances of ASL grammar and the variability in individual signing styles. Ongoing research aims to develop more sophisticated models that can adapt to these variations, ensuring accurate and fluid translations. The integration of advanced sensors, machine learning algorithms, and user-centric design continues to drive progress in this field [4].

PROBLEM DEFINITION

a. Problem Statement:

“To create a real-time system that translates spoken English into American Sign Language.”

b. Objectives:

1. To create a tool that translates English to ASL instantly for better communication.
2. To build a system that can learn and include all ASL signs, both manual and non- manual.
3. To improve learning and support for the deaf community with the translation system.
4. To help deaf people get information and services in American Sign Language (ASL).

REQUIREMENT ANALYSIS

a. Dataset Description :

- ASL dataset from Kaggle.
<https://www.kaggle.com/datasets/ayuraj/asl-dataset>
- This dataset contains 35 different directories which includes 1-9 and A-Z with their corresponding visuals.

b. Hardware Requirements :

- A computer with microphone access.
- Internet connection (for Gemini API).
- 2GB RAM or above for smooth browser rendering.
- Modern web browser (Chrome/Edge with Web Speech API support).

c. Software Requirements :

- **React.js** : for building the frontend interface.
- **Node.js & npm** : for local development and dependency management.
- **Axios** : for API communication with Google Gemini.
- **Framer Motion** : for smooth animation of ASL signs.
- **Browser API** :
 - **MediaRecorder** : For recording audio in real time.
 - **Web Audio API** : For extracting and analyzing audio features.
 - **SpeechRecognition** : For converting speech into live text.
- **Google Gemini API** : Used for punctuation correction and sentiment/emotion analysis of the spoken text.

d. Libraries Used :

- **Recharts** : To create audio feature charts (frequency, amplitude).
- **Lucide-react** : For using icons like Mic and Thermometer.
- **Axios** : For making API calls (e.g., OpenAI punctuation).
- **SpeechRecognition (Web API)** : For converting speech to text.
- **AudioContext (Web Audio API)** : For analysing and processing audio.
- **framer-motion** : For animating sign language transitions.
- **require.context (Webpack)** : For dynamically loading sign images.

SYSTEM DESIGN & IMPLEMENTATION

a. System Architecture :

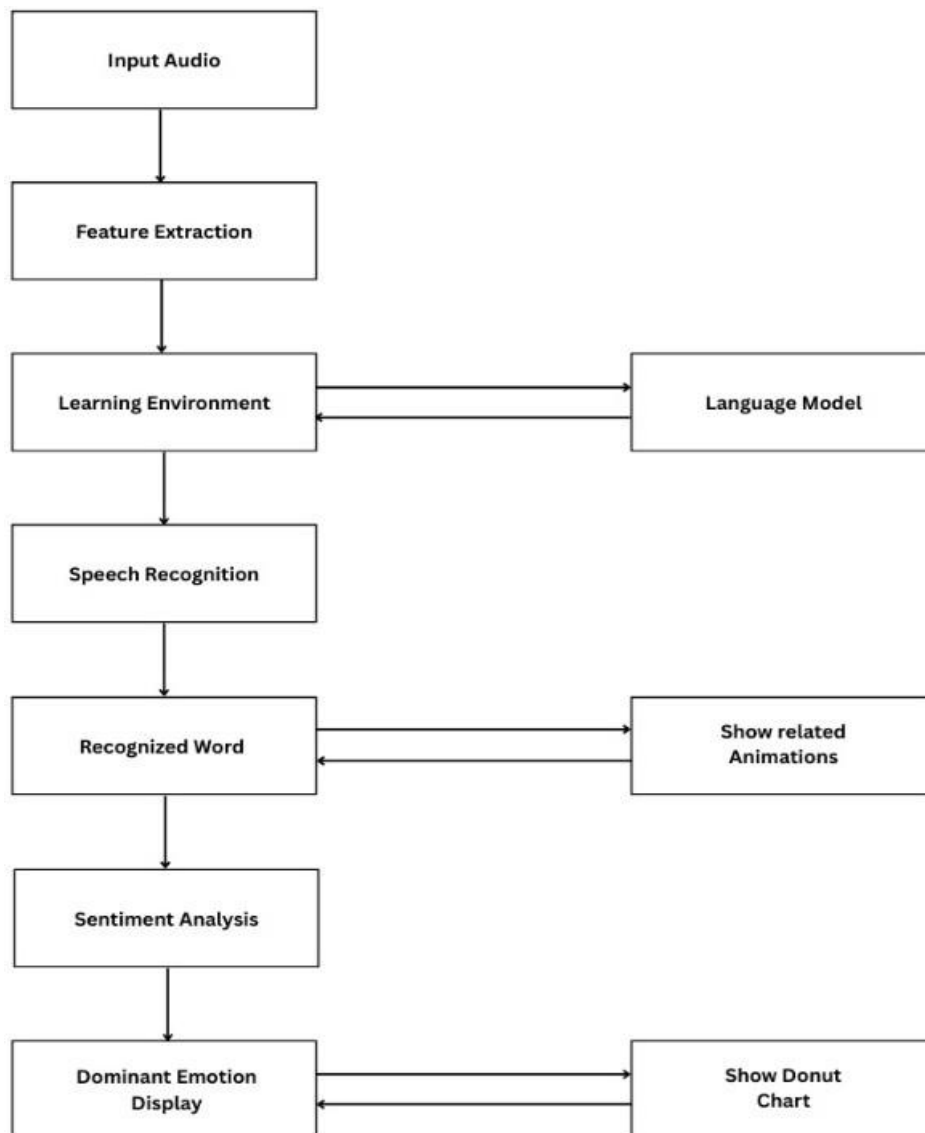


Fig. Block diagram of proposed system

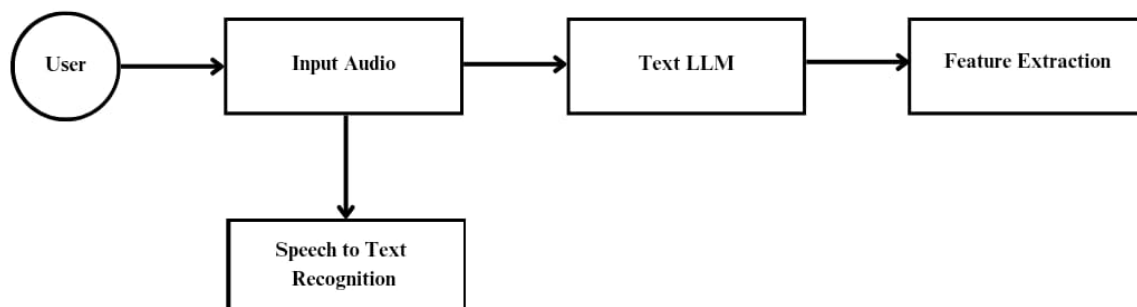
b. Data Flow Diagrams :

○ DFD Level 0 :



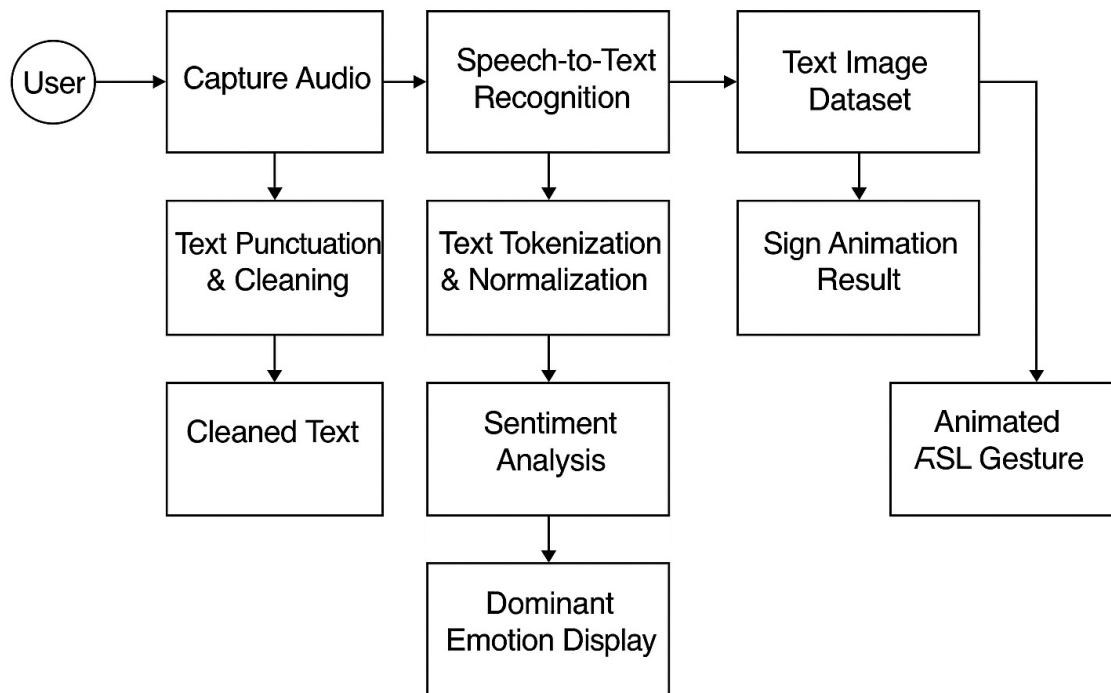
The diagram illustrates a basic audio analysis system. The user provides input audio, which is then processed through a feature extraction stage.

○ DFD Level 1 :



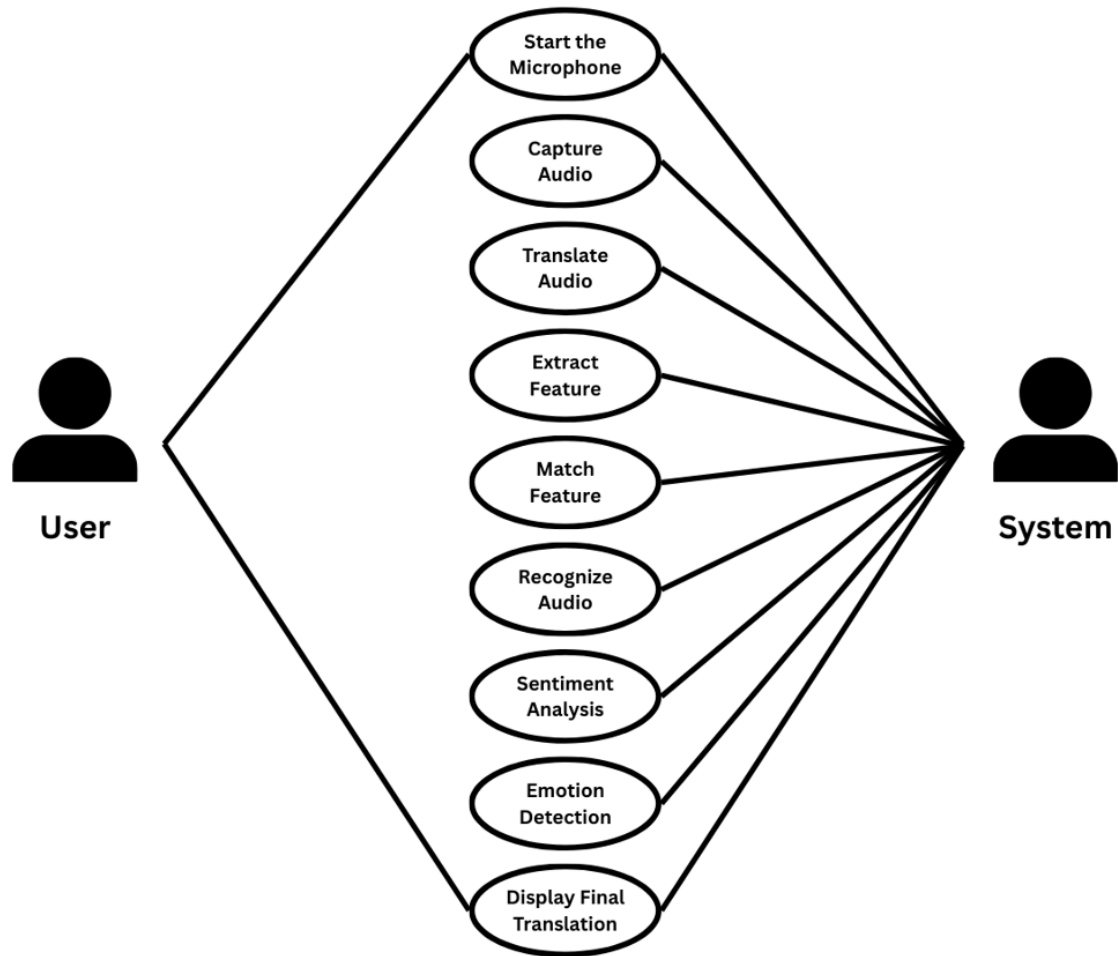
The diagram outlines a more detailed audio analysis system. The user provides input audio, which is first converted to text using a Speech-to-Text Recognition module. The resulting text is then processed by a Text LLM (likely a Large Language Model) before undergoing feature extraction.

○ **DFD Level 2 :**



The Level 2 DFD shows how user audio is captured, converted to text, cleaned, and tokenized; the text is then analyzed for sentiment to display dominant emotion and simultaneously used with a text image dataset to generate animated ASL gestures.

c. Use Case Diagram



d. Implementation Details :

1. Speech Input Processing

- **Audio Input:** Captures live audio from the user's microphone via the browser using MediaRecorder and Web Audio API.
- **Speech Recognition:** Uses built-in browser support (SpeechRecognition / webkitSpeechRecognition) for converting spoken input to real-time text.
- **Text Normalization:** A punctuation service powered by Google Gemini API is used to clean, punctuate, and capitalize the transcribed text.

2. Text Processing for Sign Translation

- **Tokenization:** Splits the cleaned text into individual words using regex and JavaScript's .split() method.
- **Simplification:** Removes punctuation and converts text to lowercase; filler words are not explicitly removed but could be implemented.
- **Translation:** Each word is broken into letters and each letter is mapped to corresponding ASL images from the dataset.

3. Sign Language Representation

- **Word-to-Sign Mapping :** Letters of each word are dynamically linked to images loaded from an ASL dataset stored in ../assets/assets/asl_dataset/.
- **Gesture Animation (optional) :** Uses framer-motion to animate transitions between letters for smooth visual presentation.
- **Sequence Animation :** The SignLanguageAnimator component handles timed, letter-by-letter display of corresponding signs.
 - Breaks the text into words and letters.
 - Animates each letter one by one using Framer Motion.
 - Displays one image per letter from the ASL dataset or a “?” placeholder if an image for the letter is missing.

4. Deployment

- **Frontend :** Built using React.js for an interactive user interface with support for animations and audio processing.
- **Backend :** Minimal - only used to connect with Google Gemini API for text correction via axios.
- **Model Optimization :** Not required since no ML model is deployed locally; entire flow is handled through browser-side logic.

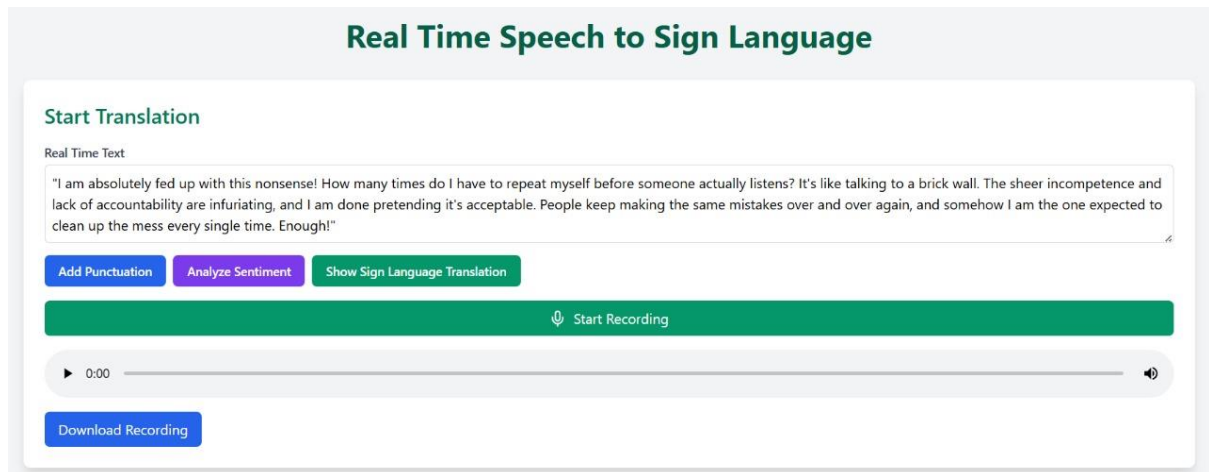
5. Visualization of Signs

- **Static** : Displays static ASL alphabet images per letter using randomized images for each letter from the dataset.
- **Dynamic** : Uses Framer Motion to create animated, smooth transitions between letter signs.
- **Emotion Visualization** : Emotion icons from Lucide-React are displayed based on the sentiment analysis results of the transcribed text.

6. Tools & Libraries

- **Speech** : Browser's Web Speech API (SpeechRecognition).
- **Text** : Google Gemini API (for punctuation and sentiment analysis).
- **Gesture** : Framer Motion (for letter animation).
- **Emotion** : Lucide-React (for displaying emotion icons based on sentiment).
- **Deployment** : Browser-based app with React frontend and API-powered text & emotion processing.
 - React is used to build a dynamic and animated UI.
 - Axios is used to connect to the Gemini API for text and emotion processing.
 - Recharts is used to show sentiment and audio data visually.
 - Framer Motion is used to animate the sign language transitions.
 - Lucide-react is used to display emotion icons based on sentiment.

RESULT ANALYSIS



This image showcases the input interface where users can provide audio input in two ways :

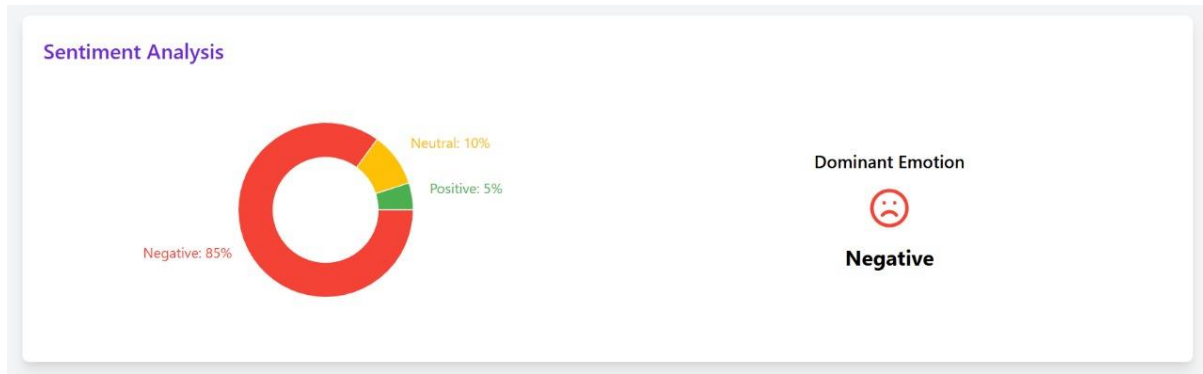
- The “Start Recording” button enables users to record their voice in real time.
- Alternatively, users can manually type the text into the "Real Time Text" input box. This dual-mode input flexibility ensures better accessibility and ease of use depending on the user's preference.

Additionally, three functional buttons have been added to enhance user control:

- “Add Punctuation” button processes the raw text using the Gemini API to add punctuation and capitalization.
- “Analyze Sentiment” button detects the emotional tone of the text and displays sentiment scores.
- “Show Sign Language Translation” button converts the processed text into animated ASL signs.

Users can also:

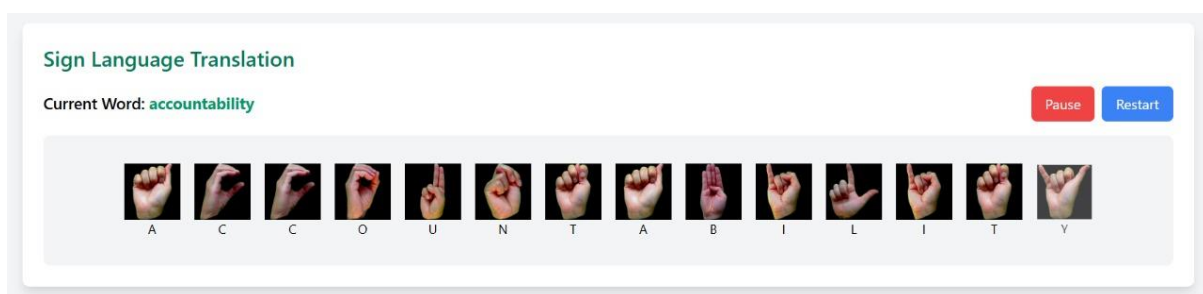
- Listen to the Recorded Voice by clicking the play button after recording.
- Download the Audio by using the provided “Download Recording” button to save the recorded file.



This image showcases the sentiment analysis features of the application :

- The “Analyze Sentiment” button detects the emotional tone of the transcribed text and categorizes it into positive, negative, or neutral.
- On the left side, a donut chart visually represents the sentiment distribution with three categories :
 - Positive (green)
 - Negative (red)
 - Neutral (yellow)
- On the right side, the dominant emotion is displayed using a simple emoji icon, representing the overall emotion detected in the text.

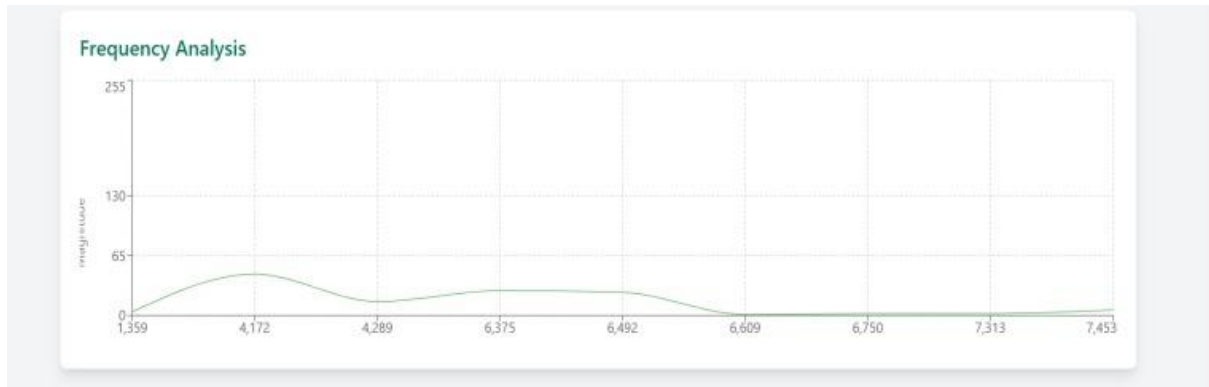
This analysis provides users with a clear, visual representation of the emotional context of the speech.



This image displays an animation of sign language gestures corresponding to the text entered or spoken:

- Each word from the input is translated into a visual sign language representation letter by letter.
- The animation shows one word at a time, allowing users to see the complete translation before moving to the next word.

- This mechanism helps users visually understand how each part of the text is signed, creating an intuitive connection between language and signs.



This image provides a detailed frequency spectrum analysis of the recorded voice.

- It shows how different frequencies are distributed in the audio signal.
- This analysis is crucial for understanding pitch, tone, and overall audio quality, and can be used in further signal processing or classification tasks.



This image presents both spectral features and time-based audio analysis.

- At the top, a waveform graph shows the amplitude over time, illustrating how loudness varies as the audio progresses.
- Below that, it extracts spectral centroid and spectral bandwidth, which are key features in analyzing the tone and clarity of the audio.

CONCLUSION & FUTURE SCOPE

The speech-to-sign language translation system enhances communication for hearing and speech-impaired individuals by converting spoken input into sign language. It combines speech recognition, real-time text, and visual sign output to create an inclusive tool. Additionally, audio analysis using spectral and time-domain features improves voice understanding, making the system both accessible and technically robust.

Future Scope :

1. Edge-Optimized ASR with On-Device LLMs :

- Replace the browser's Web Speech API with a quantized Whisper.cpp or Distil-Whisper variant for fully local transcription.
- Reduce latency and improve privacy by deploying on-device language models for punctuation and intent classification.

2. Sign Language Generation via Pose Estimation Models :

- Integrate transformer-based models (e.g., SignAll, SignFusion) that convert text to skeletal pose sequences.
- Extend the pipeline to use diffusion-based avatar synthesis, generating realistic signing avatars in real time.

3. Domain-Specific LLM Fine - Tuning :

- Fine-tune Gemini or an open LLM (like Phi-3, Mistral) for context-aware punctuation restoration using speech corpora (e.g., TED-LIUM, LibriSpeech).
- Incorporate prompt engineering with few-shot examples for domain adaptability, such as in medical or legal settings.

4. Multimodal Sentiment and Emotion Classification :

- Fuse audio features (spectral and prosodic) and LLM-transcribed text to infer speaker emotion, sentiment, or urgency.
- Apply multimodal fusion transformers or late-fusion CNN+Transformer hybrids.

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PUBLICATION DETAILS